



VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2013
Higher 1

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CANDIDATE NAME

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CLASS

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INDEX NUMBER

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BIOLOGY

8875/02

Paper 2 Core Paper

16 September 2013

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your CLASS/ INDEX no. and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Write your answers on separate answer paper provided.

At the end of the examinations,

1. Hand in section A and the question you attempted from section B separately;
2. Fasten all your work securely;
3. Enter the number of the section B question you have answered in the grid opposite.

The intended number of marks is given in brackets [] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
Total	

This paper consists of **10** printed pages.

Section A

Answer **all** the questions in this section.

- 1 Scientists discovered a novel peptide that has antimicrobial properties from a plant. This antimicrobial peptide (AMP) acts like natural antibiotic, protecting the plant from bacterial infection. **Fig. 1.1** shows the structure of this novel AMP.

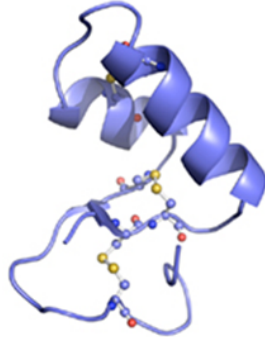


Fig. 1.1

http://www.frontiersin.org/antimicrobials_resistance_and_chemotherapy/10.3389/fmicb.2011.00232/full

- (a) Using **Fig. 1.1**, explain the different levels of organization shown by the AMP.

.....

.....

.....

.....

.....

.....

[3]

- (b) State **one** important function of the following structures in the synthesis of AMP.

(i) *nucleolus*:

.....

(ii) *nuclear pores*

.....

(iii) *golgi apparatus*

.....

[3]

When this AMP was introduced into animals, it was found that it is quickly broken down by protease enzymes present in the animal's gut. Scientists then created a man-made AMP made up of amino acids containing fluorine. This synthetic AMP is found to be more effective in killing bacteria than AMP without fluorine.

- (c) Explain clearly why AMP made from amino acids containing fluorine is more effective in killing the bacteria.

[2]

To determine the effect of this synthetic AMP, different concentrations of the AMP was introduced into bacteria suspensions and the suspensions incubated for 48 hours. The number of living bacteria were then determined and the results obtained shown in **Fig 1.2** below.

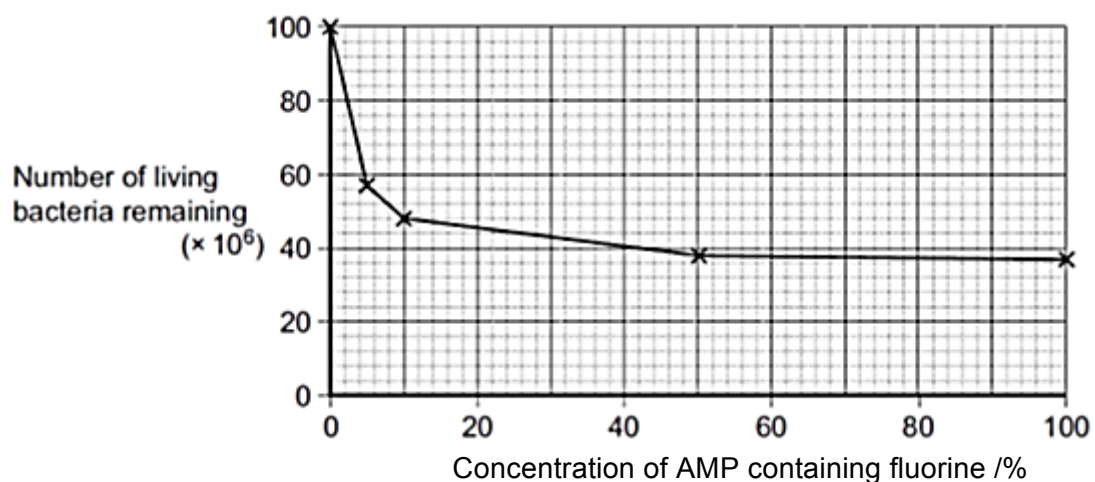


Fig 1.2

- (d) (i) Suggest an explanation for the results seen when the AMP concentration used is greater than 50%.

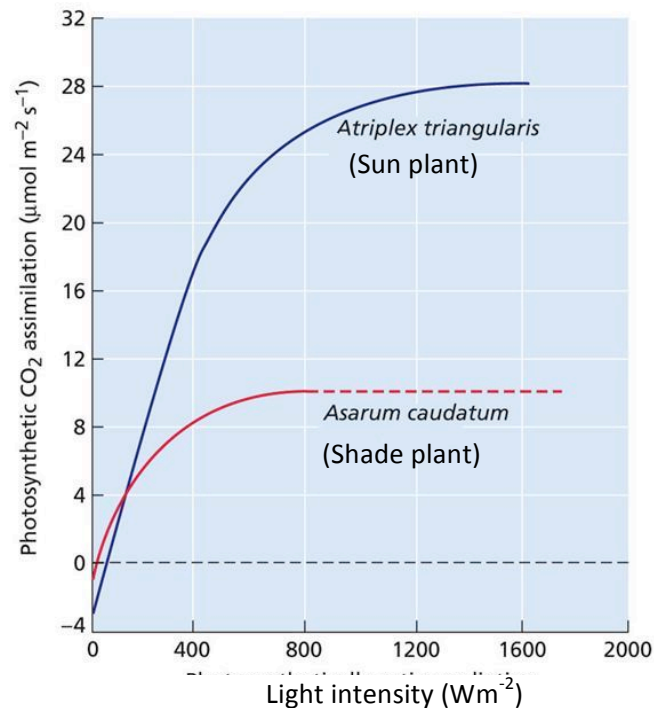
[1]

- (ii) Discuss the application of the results obtained in this experiment in the control of bacteria growth in animals.

[2]

[Total: 11 marks]

- 2 In an experiment to determine how different plants behave under varying light conditions, a plant grown in the shade (shade plant) and a plant grown in the open (sun plant) were placed under the same conditions in the laboratory. These plants were subjected to different light intensities and the amount of carbon dioxide taken up was recorded. **Fig. 2.1** shows the results obtained for these two plants.



PLANT PHYSIOLOGY, Fourth Edition, Figure 9.8 © 2006 Sinauer Associates, Inc.

<http://www.studyblue.com/notes/n/plant-phys-studying/deck/6567440>

Fig. 2.1

- (a) Using either the graph for the sun plant or the shade plant, explain the meaning of the term "limiting factor".

[2]

- (b) With reference to **Fig. 2.1**, describe two differences in the responses shown by the sun plant and the shade plant.

[2]

- (c) From the results of the experiment, suggest why plants growing in shade on the forest floors are able to survive.

[2]

- (d) The above experiment was conducted in 0.40% of CO_2 concentration. In **Fig. 2.1**, sketch a graph to show the CO_2 uptake of the sun plant when it is grown in a CO_2 concentration of 0.04%.

Label the graph C. [1]

- (e) **Fig. 2.2** shows two polymers that can be found in the sun plant.

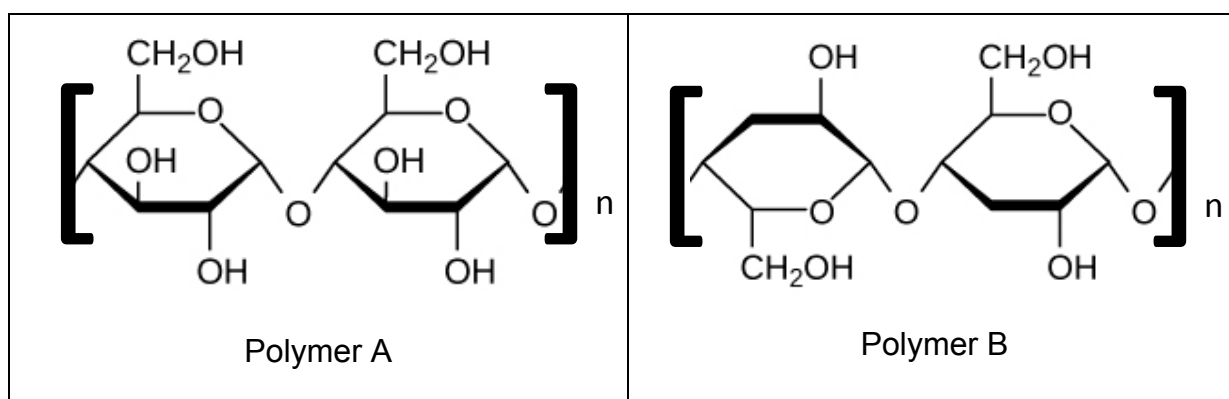


Fig. 2.2

- (i) State two differences in terms of structure between the two polymers.

[2]

- (ii) Explain how one stated structural difference in (c) (i) can be related to the different function each polymer serves in the plant.

[2]

[Total: 11 marks]

- 3 In certain breeds of domestic fowl, the loci controlling feather colour and feather shape are found on two separate chromosomes.

Feather colour can be black or barred (black with white bars) and feather shape can be straight, mildly frizzled and frizzled. In birds, females are the heterogametic sex. A breeder carried out crosses between fowls of different characteristics as shown in **Table 3**.

Cross	Parents		Progeny	
	male	female	male	female
1	barred and straight	barred and straight	all barred and straight	all barred and straight
2	barred and straight	barred and frizzled	all barred and mildly frizzled	$\frac{1}{2}$ barred and mildly frizzled, $\frac{1}{2}$ black and mildly frizzled
3	black and frizzled	black and straight	all black and mildly frizzled	all black and mildly frizzled
4	black and mildly frizzled	barred and mildly frizzled	$\frac{1}{4}$ barred and straight, $\frac{1}{4}$ barred and frizzled, $\frac{1}{2}$ barred and mildly frizzled	$\frac{1}{4}$ black and straight, $\frac{1}{4}$ black and frizzled, $\frac{1}{2}$ black and mildly frizzled
5	barred and mildly frizzled	black and mildly frizzled	$\frac{1}{4}$ barred and straight, $\frac{1}{4}$ barred and frizzled, $\frac{1}{2}$ barred and mildly frizzled	$\frac{1}{4}$ barred and straight, $\frac{1}{4}$ barred and frizzled, $\frac{1}{2}$ barred and mildly frizzled
6	barred and frizzled	black and frizzled	$\frac{1}{2}$ barred and frizzled, $\frac{1}{2}$ black and frizzled	$\frac{1}{2}$ barred and frizzled, $\frac{1}{2}$ black and frizzled

Table 3

- (a) State what is meant by the following terms:

- (i) locus (plural: loci)

[1]

- (ii) heterogametic sex

[1]

(b) Using evidence from the **Table 3**, determine the interaction between the alleles at each gene locus. Explain clearly the cross that led you to your conclusions.

[3]

(c) Using suitable symbols, draw a genetic diagram to explain Cross 4. [5]

[Total: 10 marks]

- 4 Tomatoes, when ripened will get soft and are more susceptible to damage in the process of being transported to the markets to be sold. Softening of the fruit is due to the production of an enzyme polygalacturonase (**PG**) which breaks down the cell wall. Scientists have come up with an antisense technology to block the synthesis of this enzyme so that the fruits can remain firm. This involves the insertion of a promoter region into the non-template strand of the DNA shown in **Fig. 4.1**



Fig 4.1

- (a) (i) On **Fig 4.1**, label clearly the 5' and 3' ends of both strands of DNA. [1]

(ii) Explain what is meant by the term “template”.

[1]

- (b) Explain why the insertion of a promoter in the non-template DNA strand can block the synthesis of the PG enzyme.

[3]

- (c) Describe how the promoter sequence can be isolated and then inserted into the non-template DNA.

[3]

- (d) Antisense technology has also been used in cancer therapy. Using your knowledge of cell cycle and cancer, suggest one way in which the antisense technology can be used to control cancer.

[1]

- (e) Molecular medicine is a new field of research and clinical medicine that demonstrates how information from the Human genome Project can be applied to improve human health.

Briefly describe, how information of DNA sequences can be used in molecular medicine.

[2]

- (f) Outline one ethical and one social concern arising from the Human Genome Project.

[2]

[Total 13 marks]

Section B

Answer EITHER 5 OR 6

Write your answers on the separate answer paper provided.
Your answers should be illustrated by clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

Either

- 5** (a) Describe the evidence for semiconservative replication of DNA. [6]
- (b) Explain the importance of membranes in oxidative phosphorylation. [8]
- (c) Discuss how anatomical and embryological homology can be used to support Darwin's theory of natural selection. [6]

Or

- 6** (a) Describe the mode of action of enzymes. [8]
- (b) Compare and contrast stem cells with cancer cells. [6]
- (c) Explain, using a named example, how the environment can lead to a change in the phenotype and its subsequent increase in the population. [6]

Question 4

Diabetes mellitus (Type I) is a genetic disease caused by the absence of the insulin-producing beta cells (or non-functional) of the islets of Langerhans in the pancreas, thus resulting in insulin deficiency. Conventional treatment of this disease includes standard injection of insulin which was isolated from pancreases of slaughtered pigs & cattle. Both proteins differ in one amino acid as shown in **Fig 1.1** below

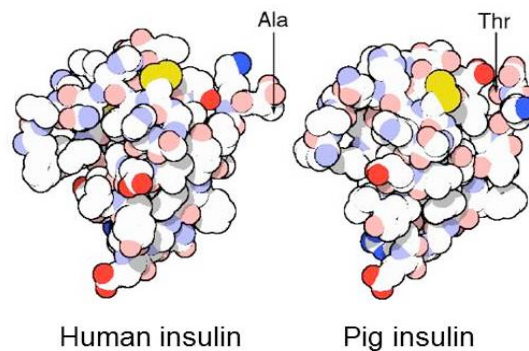


Figure 1.1

- a) Treatment of insulin isolated from pancreases of slaughtered pigs & cattle has encountered problems when injected into patients. Suggest a reason for this. [1]
- this minor difference in amino acid sequence between animal (threonine) and human insulin (alanine) allows the protein to be recognised as foreign by patient's body → immune response → degrade protein;;
- b) Genetically engineered bacteria are used in large-scale production of human insulin. **Explain the source and choice of nucleic acid to be used as template** for producing human insulin in bacteria. [3]
- **Source:** human pancreas/ beta cells of islets of Langerhans express insulin gene
 - **Type of nucleic acid as template:** mRNA is extracted from this pancreas/beta cells and used as a template for the synthesis of complementary DNA (cDNA) {by the enzyme reverse transcriptase}.
 - **Advantage:** prokaryotes lack post-transcriptional machinery/ do not have mRNA processing steps and mRNA template will not have introns which requires post-transcriptional processing/intron splicing
- c) (i) In human beta pancreatic cells, insulin is initially synthesized as preproinsulin. **Preproinsulin is a precursor molecule that consists of 110 amino acids, which** includes a 24 amino acid long signal peptide. The signal peptide requires post-translational processing, an event that occurs at the rough endoplasmic reticulum. Insulin is commercially produced in bacteria. Suggest how DNA gel electrophoresis can be used to confirm that genetically engineered bacteria cells contain the insulin gene after positive antibiotic selection. [3]
- Extract DNA from clones of bacteria that survive on antibiotic medium
 - RE digest to release GOI from recombinant DNA
 - Gel electrophoresis separate vector and insert DNA according to mw
 - Expect a $(86 \times 3) + 3$ bp fragment on agarose gel that is insulin cDNA/DNA band

(ii) Suggest how stem cells can be used for long term treatment of Type I diabetes mellitus. [4]

- *Ex vivo* treatment by isolating hematopoietic stem cells from bone marrow of patient
- Replace mutant allele of insulin gene with normal copy coding for functional protein
- Expand recombinant stem cells that contain the insulin cDNA
- by asymmetric division Induce HSCs to transdifferentiate into differentiating cells of pancreatic cell lineage (*application of Fig 1, p5 with transdifferentiation concept on p5*)
- Reintroduce back into patient
- generate specialised cells that express and secrete insulin
- Pool of unspecialised stem cells capable of long term self-renewal

(d) Molecular medicine is a new field of research and clinical medicine that demonstrates how information from the Human genome Project can be applied to improve human health. Briefly describe, using two examples, how information of DNA sequences can be used in molecular medicine. [2]

Any two

- Improved diagnosis of genetic diseases – gene mutations and cause / progression of diseases
- Earlier detection of genetic predisposition to disease
- Drugs designed based on molecular information (eg, inhibitors of proteins involved in disease progression)
- Customised drug design based on individual genetic profiles for greater drug efficacy
- Gene therapy to cure genetic diseases
- AVP

(e) Outline one ethical and one social concern arising from the Human Genome Project. [2]

Ethical

- Commercialization/ patents/ copyrights of genetic sequences – do companies/ labs have the right to patent DNA sequences and limit their accessibility to others for research and development into useful products
- **Fairness in use of genetic information** – Ref access and use by employers, insurers, courts, schools, adoption agencies, military, etc.
- **Privacy and confidentiality issues** – control and ownership of genetic information
- **Reproductive issues** involving informed consent and use of genetic information in reproductive decision making. Do parents receive adequate genetic counselling about risks and limitations of genetic technology? Reliability and usefulness of fetal genetic testing
- **Uncertainties associated with value of gene testing** – for complex diseases caused by multiple genes that involve gene-environment interactions
- **Clinical issues** – education of healthcare professionals, patients and general public wrt genetic capabilities, scientific limitations and social risks;
- **Clinical issues** – implementation of standards and quality control measures in testing procedures

Social

- **Philosophical implications** regarding free will, individual responsibility vs genetic determinism. Ref questions about link between genes and behaviour
- **Philosophical implications** regarding boundary between medical treatment vs enhancement
- **Physiological impact and stigmatization** – extent to which an individual's genetic makeup affects him and society's perception of the individual.

- **Physiological impact and stigmatization** – impact of genomic information on members of minority communities